PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Economic Burden of Road Traffic Injuries in Sub-Saharan Africa; A Systematic Review of Existing Literature
AUTHORS	Ryan-Coker, Marcella; Davies, Justine; Rinaldi, Giulia; Hasselberg, Marie; Marke, Dennis; Necchi, Marco; Haghparast-Bidgoli, Hassan;

VERSION 1 – REVIEW

REVIEWER	Krebs, Elizabeth Thomas Jefferson University
REVIEW RETURNED	17-Feb-2021

GENERAL COMMENTS	Thank you to all of you for this important and challenging work - I've engaged in a related review to quantify simply the instances of RTIs described in the SSA lit and it's a lot of work and frustration! I defer to others with more experience and familiarity with your databases than myself but I wonder if your search strategy could be expanded? We used PubMed, EMBASE, Africa-Wide Information, Global Health, and Web of Science. are they all included based on your approach? I also may have missed it but I think it would be helpful to know the series of reasons for excluding so many abstracts, was there any system applied to how your 2 reviewers considered each one or simply gestalt and consensus? While I think you addressed the major PRISMA points and I'm happy to see some clarity about the actual quantity and quality of the lit that's out there regarding economic burden of RTI in SSA I personally would be more comfortable with additional reassurance that these 11 you selected absolutely reflect all that's out there with enough data to contribute to be useful - understanding more precisely how/why the exclusions at each level and making sure these other databases we found useful in our own similar review were considered

REVIEWER	Wu, Shaowei Peking University , Department of Occupational and Environmental Health Sciences
REVIEW RETURNED	08-Mar-2021

GENERAL COMMENTS	This systematic review explored and synthesized existing literature on the direct and indirect costs from RTIs in Sub-Saharan Africa (SSA), the quality of existing evidence, methods used to estimate and report these costs, and the factors that drive the costs. The information presented in this study fills the gap in the existing

	literature about the economic burden of RTIs in SSA. I have a few comments to improve the manuscript as follows. In Figure 1, the reasons for exclusions at the second and third steps (n=491 and n=129) should be stated. Lines 44-48, page 9: The authors stated that “one of these studies were excluded because the same unit costs were applied for two studies in the same country- one applied at provincial and the other at national level”, the excluded study should be cited here. Additional file 3 was not cited in the text. It seems the file repeats the contents reported in Fig. 2. If so, I suggest replace Fig. 2 with the file 3, as the information reported in file 3 has better clarity compared to Fig. 2. Study limitation: the study covers only six countries in the sub-Saharan Africa area and is limited to literature published in English, whereas it is possible that reports in some countries may not be published in English and were thus omitted from the review. The study would better acknowledge this point and provided relevant statement. Lines 36-40, page 20: The study concluded that “the available evidence shows that the economic burden attributed to RTIs is substantially high in SSA” may not be appropriate, as the summarized costs were only estimated at per individual/injury level but not at the population level, and the quantitative comparison to the same costs in other areas out of SSA or to the local economic development level of the included countries was unclear. Lines 26-27, page 6: there is an extra word ‘and’ in the statement ‘with cost and or cost analysis’

VERSION 1 – AUTHOR RESPONSE

Reviewer 1-

1.Dear reviewer, Thank you for your comments. We searched all the above databases except for Africa-Wide Information. We also included ProQuest Central, Econlit, World Bank Group e-Library, Google scholar and World Health Organisation (WHO) webpage. We were able to access EMBASE, Web of Science, ProQuest Central and Econlit databases free of charge through UCL library. However, we believe that combination of all these databases have captured majority of the literature available in this area.

2.We tried to explain this briefly in Fig 1. We followed the PRISMA guidelines for screening and exclusion. We found that most of the articles that were returned in the search were not costing or cost of illness studies. We have added more explanation on the reason for excluding the abstracts in results section (page 9) which is as follows: ‘A substantial number of abstracts were removed during the screening process as they were outside the scope and/or did not relate to the themes of this review.’

3.We are confident that the studies reviewed reflect the existing literature on the topic. Here is further explanation in addition to figure one for exclusion of articles at each level:

872 articles were returned from all databases searched. 249 of these were duplicates and were removed in the initial stages. The remaining 623 articles were then screened based on titles. 491 studies were excluded. These studies were excluded because the titles may have contained one of the search terms but there were not related to the themes of this review (i.e., Cost/Economic Burden, Road Traffic Injuries and Sub-Saharan Africa). The abstracts of the remaining 132 studies and some hand-picked references were then assessed for eligibility. Articles that were epidemiological studies, unretrievable abstracts or those that were not original studies were not included for full text review. After these exclusions, the full texts of 35 studies were reviewed. The details of exclusion for these texts are shown in Fig 1.

Reviewer 2-

1.Dear reviewer, thank you for your comments. We have added more explanations on the process of exclusion to the result section on page 9 in the manuscript as explained in response to comments 2 and 3 above.

2.This study has been cited as suggested.

3.Thank you for pointing this out. Additional file 3 is Additional file 2 which was cited in the text. We have now included this in text along with Fig 2 for better clarity on page 14.

4.This has been covered in the last sentence in the imitations (on page 20) which now reads ‘Likewise, the number of studies and countries included in this literature review is quite small, owing to the lack of existing data on the topic published in English and is not entirely representative of SSA.’

5.We have revised this sentence to read ‘Considering high and increasing rate of RTI in SSA, the available evidence suggests that the economic burden of RTIs in SSA can be substantially high.’

6.The extra ‘and’ has been deleted.